

Business Requirements Document

SOC Operations Dashboard — Power BI

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1. Executive Summary

This document defines the business requirements for a Power BI Security Operations Center (SOC) dashboard consolidating data from FortiSOAR alert management, Tenable vulnerability scanning, and threat intelligence feeds. The dashboard provides two distinct user experiences — an executive view for strategic monitoring and an operational view for analyst-level investigation — both backed by a star-schema data model and DAX measures.

The deliverable demonstrates the candidate's ability to ingest heterogeneous security data, design a dimensional model, author analytical measures, and present operational intelligence in a manner consistent with enterprise SOC reporting standards.

2. Background and Context

2.1 Purpose

Modern Security Operations Centers generate large volumes of telemetry across disparate systems: SOAR platforms for alert handling, vulnerability scanners for asset risk posture, and threat intelligence feeds for known-malicious indicators. Each system answers different questions, and consolidating them into a unified view enables both strategic decision-making (by leadership) and operational triage (by analysts).

This dashboard addresses that consolidation need within Power BI, demonstrating end-to-end SOC analytics capability from raw JSON ingestion through to a finished, brand-aligned deliverable.

2.2 Scope

In scope:

- Ingestion and transformation of three SOC data sources within Power BI (FortiSOAR alerts, Tenable vulnerabilities, threat intelligence)

- Construction of a star-schema dimensional model
- Authoring of DAX measures supporting the prescribed KPIs (MTTD, Alert Volume by Severity, Escalation Rate) and additional analytical metrics
- Design and implementation of an executive-level dashboard view
- Design and implementation of an operational-level dashboard view with interactive filtering
- Documented data quality findings and modeling assumptions

Out of scope:

- Integration with the AbuseIPDB enrichment file (not provided in the data package — see Section 7.2)
- Real-time data streaming or scheduled refresh from live source systems
- User access management, row-level security, or workspace deployment
- Mobile-optimized layouts (desktop-first design)
- Alerting, paging, or external notification integration

2.3 Stakeholders

Stakeholder	Role	Primary use
CISO / Security Leadership	Strategic oversight	Executive view: weekly health check, board reporting
SOC Manager	Operational oversight	Both views: workload visibility, analyst coordination
SOC Analyst (Tier 1/2)	Day-to-day operations	Operational view: filtering, triage, IOC investigation
Vulnerability Management	Risk remediation	Operational view: asset hotspots, CVE prioritization
Threat Intelligence Team	Intel coverage assessment	Operational view: IOC overlap analysis

3. Business Requirements

3.1 Functional Requirements

ID	Requirement
FR-01	The dashboard shall display the total number of alerts ingested.
FR-02	The dashboard shall display alert volume broken down by severity (Critical, High, Medium, Low).

- FR-0 3 The dashboard shall display the Mean Time to Detect (MTTD) as a calculated metric.
- FR-0 4 The dashboard shall display the alert escalation rate as a calculated metric.
- FR-0 5 The dashboard shall display average alert resolution time.
- FR-0 6 The dashboard shall display open vulnerability counts broken down by severity.
- FR-0 7 The dashboard shall display a remediation rate metric for vulnerabilities.
- FR-0 8 The dashboard shall provide a time-series visualization of alert volume.
- FR-0 9 The dashboard shall provide at least one executive-level view summarizing KPIs without requiring user interaction.
- FR-1 0 The dashboard shall provide at least one operational-level view supporting interactive filtering.
- FR-1 1 The operational view shall allow filtering by date range, severity, status, alert category, and threat type.
- FR-1 2 The operational view shall enable cross-visual filtering (clicking a chart element filters related visuals).
- FR-1 3 The dashboard shall display the top N highest-risk vulnerabilities sorted by CVSS score.
- FR-1 4 The dashboard shall display the top IOCs by alert frequency, with available threat intelligence enrichment.
- FR-1 5 The dashboard shall provide a mechanism to clear all applied filters in a single action.
- FR-1 6 The dashboard shall display the data refresh timestamp.
- FR-1 7 The dashboard shall provide page-level navigation between executive and operational views.

3.2 Non-Functional Requirements

ID	Requirement	Acceptance criteria
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NFR-01	All data preparation must be performed within Power BI (Power Query M and DAX).	No external ETL tooling used; source files referenced directly.
NFR-02	All KPIs must be implemented as DAX measures, not calculated columns.	Visible in the Measures table; no calculated columns serving as KPI sources.
NFR-03	The data model must follow star-schema design principles.	Fact tables surrounded by dimension tables; no flat or snowflake structures except where conformed dimensions naturally span fact tables.
NFR-04	Visual styling must be consistent across both pages.	Unified palette, typography, spacing, and chrome between executive and operational views.
NFR-05	Severity color encoding must be consistent across all visuals.	Critical → red, High → orange, Medium → yellow, Low → green throughout.

3.3 Data Requirements

ID	Requirement	Notes
DR-01	Source: <code>fortisoar_alerts.json</code> (2,000 records, JSON-LD format)	FortiSOAR SOAR alert export
DR-02	Source: <code>tenable_vulnerabilities.json</code> (2,000 records, flat array)	Tenable vulnerability scanner output
DR-03	Source: <code>threat_intelligence.json</code> (2,000 records, flat array)	Internal threat intelligence feed
DR-04	Source: <code>abuseipdb_enrichment.json</code> (2,000 records expected)	Not received — see Section 7.2
DR-05	Alert records must retain: identity, severity, status, timestamps, owner, category, and indicators of compromise (IOCs)	All fields preserved through transformation

DR-06	Vulnerability records must retain: asset identifier, CVE, severity, CVSS score, first-seen and last-seen dates, status	All fields preserved
DR-07	Threat intelligence records must retain: IOC value, IOC type, threat type, confidence, date range, source	All fields preserved
DR-08	The model must support cross-source analysis (alerts joined to threat intelligence via IOCs)	Bridge-table pattern used

4.2 Two-View Design Rationale

View	Audience	Design intent
Executive Overview	CISO, board, leadership	Glanceable summary. No filters. Eye flow: KPI strip → trend → breakdowns. Reads in 60 seconds.
Operational Detail	SOC analysts, vulnerability managers	Filterable workspace. Persistent slicer panel. Dense data tables for triage. Designed for daily analyst use.

The two views answer fundamentally different questions and are therefore structured differently, while sharing visual language (palette, typography, layout grid) for consistency.

4.3 KPI Inventory

Category	KPI	Source data	Measure name
Volume	Total Alerts	FactAlerts	Total Alerts
Volume	Alerts by Severity	FactAlerts × DimSeverity	Critical Alerts, High Alerts, etc.
Operations	MTTD (Mean Time to Detect)	FactAlerts (ingested – created)	MTTD (min), MTTD Display
Operations	MTTR (Avg Resolution Time)	FactAlerts (updated – created)	MTTR (min), MTTR Display
Operations	Escalation Rate	FactAlerts × DimSeverity × DimStatus	Escalation Rate
Workflow	Open / Closed Alerts	FactAlerts × DimStatus	Open Alerts, Closed Alerts
Workflow	Closure Rate	FactAlerts × DimStatus	Closure Rate

Risk	Open Critical Vulnerabilities	FactVulnerabilities × DimSeverity	Open Critical Vulns
Risk	Remediation Rate	FactVulnerabilities	Remediation Rate
Intelligence	Unique IOCs (in alerts)	FactAlertIOCs	Distinct Alert IOCs
Intelligence	TI Coverage Rate	DimIOC	TI Coverage Rate

5. User Experience

5.1 Executive Overview Page

Zone	Content	Purpose
Header	Title bar, reporting period	Context
KPI strip (top)	5 cards: Total Alerts, % High/Critical, Escalation Rate, Avg Resolution Time, Open Critical Vulns	Headline metrics
Trend chart (middle)	Time-series of alert volume	Temporal context
Severity donut (bottom-left)	Volume by severity	Risk distribution
Vulnerability bar chart (bottom-right)	Top 10 assets by open vulnerabilities	Risk hotspots

5.2 Operational Detail Page

Zone	Content	Purpose
Header	Title bar, last refreshed timestamp	Context
Filter rail (left)	Date, Severity, Status, Category, Threat Type slicers + clear button	Analyst-driven filtering
KPI strip (top)	4 cards: Open Alerts, Critical Alerts, MTTD, Unique IOCs	Operational headlines
Stacked bar (middle-left)	Alerts by category, segmented by severity	Workload composition
100% stacked column (middle-right)	Alert pipeline (Open/In Progress/Closed) by severity	Workflow status

IOC table (bottom-left)	Top IOCs by alert frequency, with TI enrichment	Threat-surface scan
Vulnerability table (bottom-right)	Open vulnerabilities sorted by CVSS	Patch prioritization

5.3 Interaction Model

- All filters in the slicer rail apply globally to the operational page.
- Cross-visual filtering enabled: clicking a chart element filters related visuals.
- Visuals unrelated to a clicked element remain unaffected (e.g., clicking an IOC row does not filter the vulnerability table).
- The "Clear all filters" button resets the page via a Power BI bookmark.

6. Data Modeling Approach

6.1 Star Schema Decision

The model implements a star schema with three fact tables and a bridge table. Rationale:

- **FactAlerts** captures alert events (one row per alert).
- **FactVulnerabilities** captures vulnerability findings (one row per finding).
- **FactAlertIOCs** is a bridge between alerts and indicators, enabling many-to-many resolution.
- **Conformed dimensions** (**DimSeverity**, **DimDate**) are shared across multiple fact tables, allowing cross-fact analysis (e.g., "Critical severity" filters both alerts and vulnerabilities simultaneously).

6.2 Dimension Design

Dimension	Cardinality	Notes
DimSeverity	5 rows	Manually constructed; includes sort order and color hex
DimStatus	3 rows	With IsOpen boolean flag
DimCategory	4 rows	Composite key Category + SubCategory
DimOwner	2 rows	Includes synthetic "Unassigned" record for null owners (33% of alerts)
DimDate	~850 rows	DAX-generated calendar, marked as date table
DimAsset	200 rows	Distinct asset identifiers

DimCVE	1,788 rows	Distinct CVE identifiers
DimIOC	~2,232 rows	UNION of alert IOCs and threat intel feed

6.3 DimIOC Design Note

DimIOC is constructed by union rather than direct sourcing from threat intelligence. Rationale: in real-world data, alerts contain IOCs that may not appear in any threat intelligence feed, and threat intelligence contains IOCs that may not appear in any alert. A union-based dimension preserves all distinct IOCs with TI enrichment as nullable columns, supported by an InAlerts provenance flag. This pattern enables future analysis of TI coverage rates and supports the addition of further enrichment sources (such as AbuseIPDB).

6.4 Relationship Strategy

All relationships are Many-to-One from fact to dimension, with single-direction cross-filter (from dimension to fact). Bidirectional relationships are avoided to prevent filter ambiguity in a multi-fact model.

Inactive relationships are used on FactAlerts for the IngestedDate and UpdatedDate columns, accessed via USERRELATIONSHIP() in time-context measures.

7. Data Quality and Assumptions

7.1 Synthetic Data Characteristics

The provided dataset is synthetic with several uniform distributions that affect measure variance:

Field	Observation	Impact
Severity	Exactly 500 alerts per level	KPI breakdowns are uniform; severity comparisons show no variance
Status	Approximately even split across 3 statuses	Status-based metrics show balanced distribution
Category	Exactly 500 per category	Same uniform pattern
Detection delta (ingested - created)	Constant 5 seconds across all alerts	MTTD measure returns a constant value

Response delta (updated - created)	Constant 5 minutes across all alerts	MTTR measure returns a constant value
Alert creation date	All 2,000 alerts on 2026-05-05	Time-trend visualizations show limited variance
Source system	Constant "FortiSOAR"	DimSource dimension was not created (no cardinality)
Alert type	Constant "Security Alert"	Field dropped from model
Plugin name (vulnerabilities)	Constant "Generic Vulnerability"	Field dropped from model

These characteristics are noted in the dashboard's Assumptions documentation. On real-world data, the same measures would produce meaningful variance without requiring redesign.

8. KPI Definitions

8.1 MTTD (Mean Time to Detect)

Formula: `AVERAGE(ingested_at - created_at)` in minutes **Interpretation:** Time elapsed between event occurrence and SOC ingestion. **Synthetic data result:** Constant 5 seconds (documented limitation). **Production behavior:** Variable across alerts; expected to reveal ingestion latency patterns.

8.2 Alert Volume by Severity

Formula: `COUNTROWS(FactAlerts)` evaluated within severity filter context. **Implementation:** Individual measures per severity (`Critical Alerts`, `High Alerts`, etc.) plus `% High or Critical` aggregate.

8.3 Escalation Rate

Formula: `Escalated Alerts / Total Alerts`, where Escalated is defined as Severity $\in$ {High, Critical} AND Status $\neq$ Closed.

Definition rationale: The dataset does not contain an explicit "Escalated" status value. Among alternative definitions considered (severity-only, status-only, playbook-based, score-threshold), the chosen formulation produces a semantically meaningful metric reflecting active high-priority workload. Documented in the Assumptions page; alternative definitions can be substituted with a single measure edit.

8.4 Avg Resolution Time (MTTR)

Formula: `AVERAGE(updated_at - created_at)` in minutes **Interpretation:** Time elapsed between alert creation and most recent update (proxy for resolution). **Synthetic data result:** Constant 5 minutes.

11. Acceptance Criteria

The deliverable will be considered complete when:

1. The `.pbix` file opens without errors and refreshes successfully against the provided source files.
2. All Must-priority functional requirements (FR-01 through FR-12) are demonstrably implemented.
3. All non-functional requirements (NFR-01 through NFR-07) are satisfied.
4. The two-view design demonstrates discernible differences in audience and purpose (executive vs. operational).
5. Modeling and KPI logic match the descriptions in Sections 6 and 8.
6. Visual styling is consistent across both pages per Section 9.
7. All identified data quality findings are documented in the Assumptions section of this BRD or in the dashboard itself.

Document control

Version	Date	Author	Changes
1.0	14 May 2026	[Your Name]	Initial draft
